

The dependence of the dynamic moduli on the director angle anchoring condition for heterogeneous nematic polymers.

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ABSTRACT

The storage and loss moduli for nematic liquid crystal polymers are predicted from asymptotic analysis of small amplitude oscillatory shear flow with one-dimensional spatial heterogeneity. We pay special attention to the anchoring angle of the major director at the parallel plates, which can cause a variation of two-to-three orders of magnitude in the storage modulus, among other effects. Our numerical solutions for a system consisting of the velocity, director angle, and two order parameters show an interesting combination of effects from the analytical solutions of two simpler formulations of the problem:

(1) We see Leslie-Ericksen-like behavior for normal and tangential anchoring because the order parameters are zero for these anchoring conditions. If the order parameters are ignored for other anchoring angles, the remaining system for the velocity and director angle gives a similar analytical prediction of the moduli.

(2) If the heterogeneity is ignored, we recover the monodomain model, which has analytical solutions for the order parameters with oblique anchoring angles and predictions of the moduli.

We find that a linear superposition of these two simple models can be quite effective in capturing the behavior of the moduli from the full heterogeneous system with order parameters. For both the storage and loss moduli, we found that for low frequencies and for high frequencies, the Leslie-Ericksen-like behavior is dominant. However, for a window of moderate frequencies, the monodomain contribution is dominant, but we find that this feature is dependent on the spacing between the parallel plates.